

1. Title Page

Title: Analysis of Customer Attrition Drivers in European Retail Banking: A Data-Driven Approach to Retention Strategy

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Project Context: European Banking Churn Internship Project

2. Abstract

This research investigates the primary drivers of customer churn within a major European banking institution using a dataset of 10,000 clients. By developing an interactive analytics suite using Python and Streamlit, this study identifies critical risk segments based on geography, age, and financial health. Key findings reveal a significant churn outlier in the German market (32.44%) and a high-risk demographic peak among customers aged 46–60. Furthermore, the analysis highlights a correlation between product saturation and increased attrition. These insights provide a framework for targeted loyalty interventions aimed at high-value customer retention.

3. Introduction

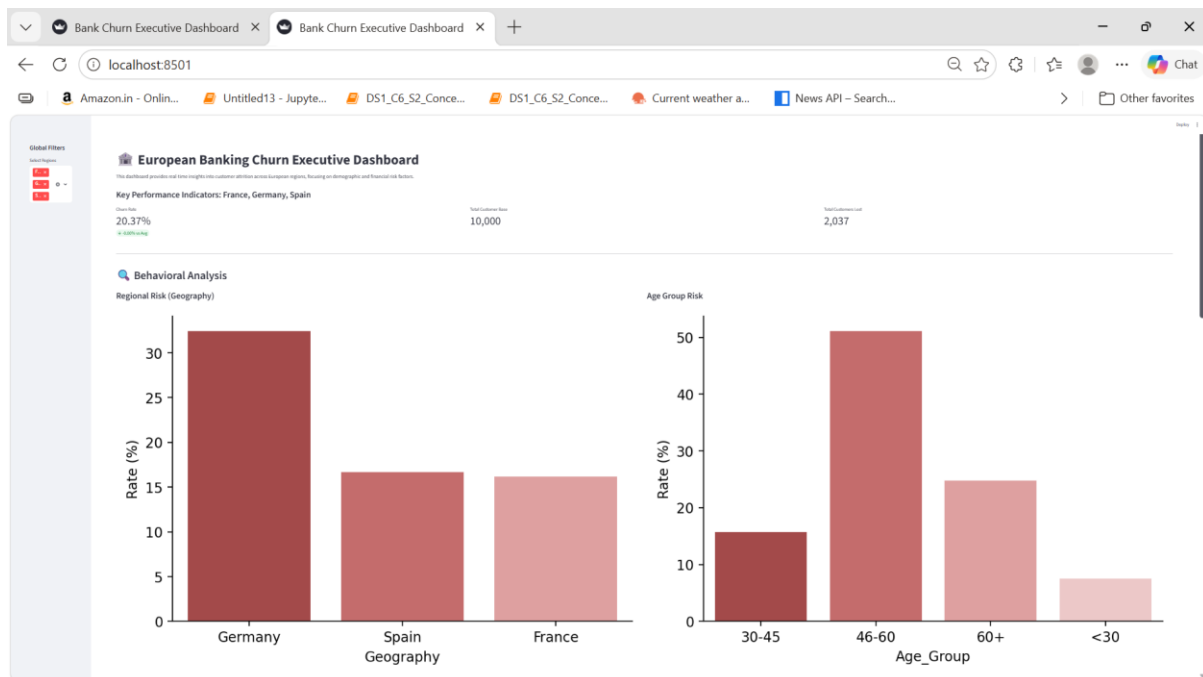
In the modern banking landscape, customer retention is a primary driver of long-term profitability. Studies suggest that acquiring a new customer is significantly more expensive than retaining an existing one. This paper explores "Churn"—the phenomenon where customers close their accounts—to identify patterns that precede an exit. By leveraging Python-based Exploratory Data Analysis (EDA), this project aims to provide stakeholders with actionable intelligence to reduce attrition rates and protect the bank's core revenue streams.

4. Methodology

This study utilized a multi-stage data science workflow to process and visualize banking churn factors:

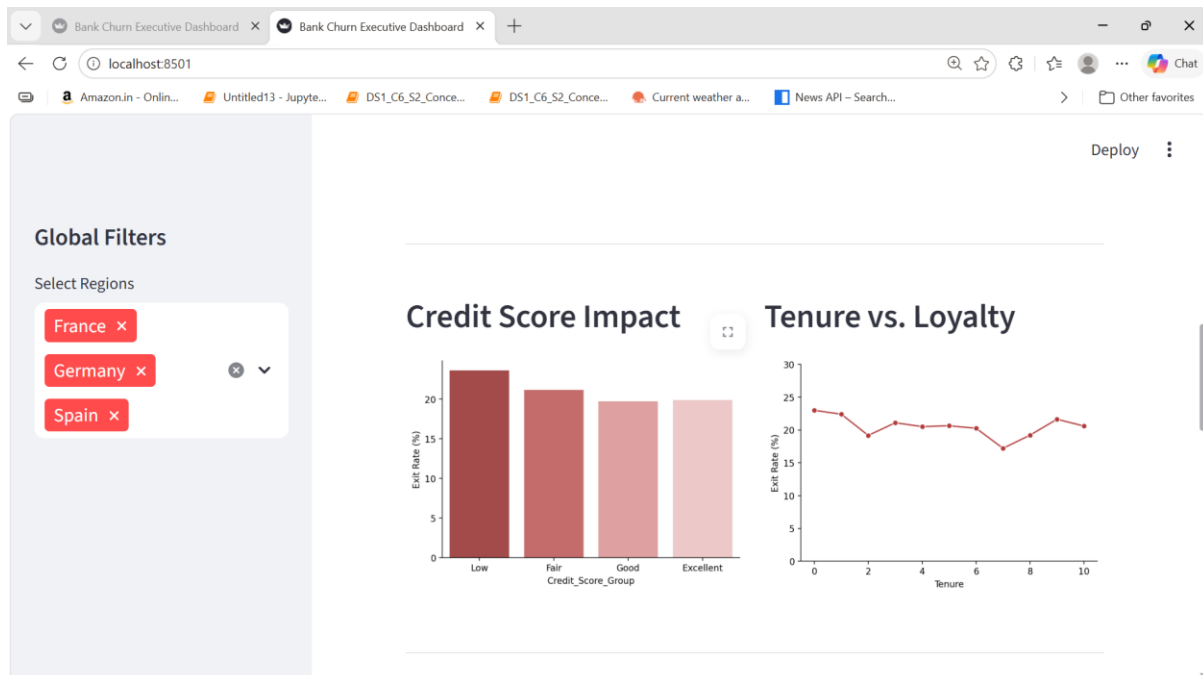
- **Data Acquisition:** A primary dataset of 10,000 customer records was utilized, containing demographic (Age, Geography, Gender) and financial (Balance, Credit Score, NumOfProducts) variables.
- **Preprocessing & Feature Engineering:** Data was cleaned using the **Pandas** library. A custom feature, `Credit_Score_Group`, was engineered to segment customers into 'Low', 'Fair', 'Good', and 'Excellent' categories to better identify financial risk thresholds.
- **Interactive Analytics Environment:** The **Streamlit** framework was employed to build a real-time executive dashboard. This allowed for dynamic filtering by region (France, Spain, Germany), enabling a granular comparison of localized churn drivers.
- **Data Visualization:** Statistical distributions and categorical risks were mapped using **Seaborn** and **Matplotlib**, employing a unified color-coded system to highlight critical attrition "hotspots."

5. Results and Discussion



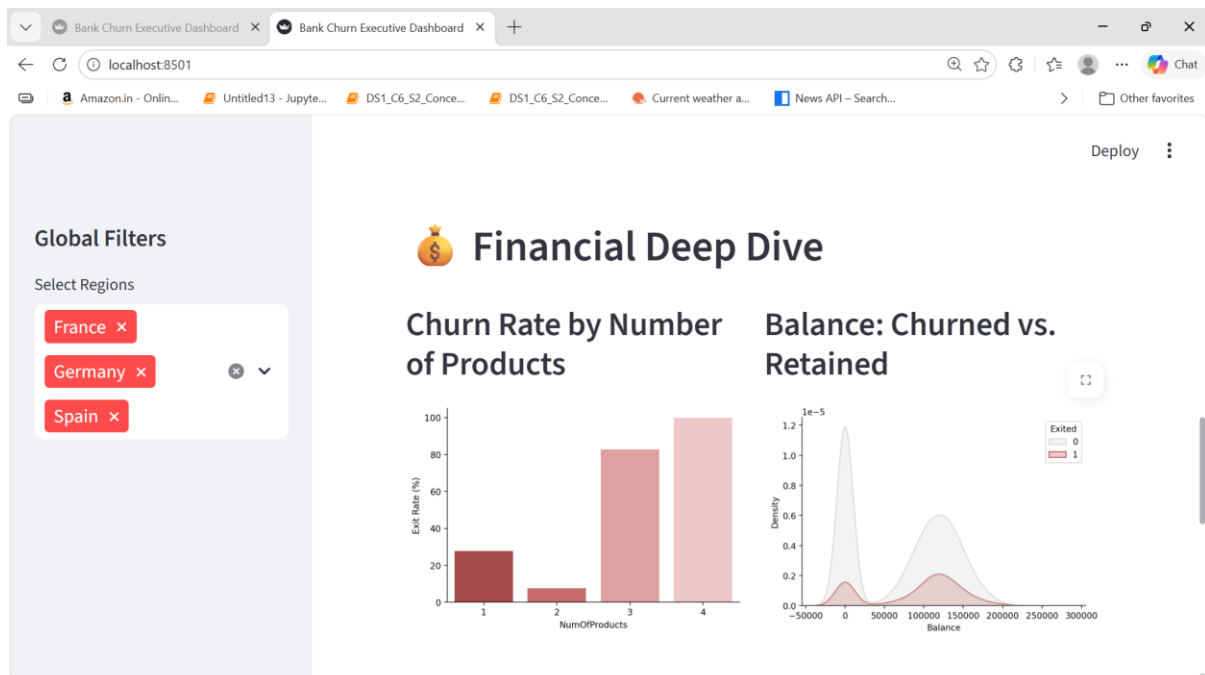
5.1 Geographic and Demographic Insights The analysis reveals a significant disparity in churn rates across European markets. Germany is identified as the highest risk region with a churn rate of approximately 32%, nearly double the rates observed in France and Spain.

Demographically, the 46–60 age group represents a "danger zone" for the institution. This segment shows the highest peak in attrition, suggesting that established customers in their peak earning years are the most likely to migrate to competitors.



5.2 Credit Risk and Tenure Analysis The analysis further examines the relationship between financial standing and customer loyalty. Key observations include:

- **Credit Score Correlation:** There is a clear inverse relationship between credit scores and churn. Customers in the 'Low' credit score bracket exhibit the highest attrition rates, suggesting a higher sensitivity to banking fees or a lack of personalized credit products for this segment.
- **Tenure Paradox:** The 'Tenure vs. Loyalty' trend indicates that churn is not strictly limited to new customers. In fact, attrition remains relatively stable across various tenure lengths, implying that the bank must focus on "continuous engagement" rather than just initial onboarding satisfaction.



5.3 Financial Deep Dive and Product Saturation The most critical findings emerge from the financial data analysis:

- **Product Saturation Risk:** A staggering trend is observed where customers with 3 or 4 products have a churn rate approaching 100%. This suggests a major failure in the service model for multi-product holders, likely due to high complexity or poor cross-channel support.
- **Balance Vulnerability:** The balance distribution density plot reveals that "Churned" customers are heavily concentrated in the mid-to-high balance range (100k - 150k). This represents a significant loss of liquidity for the institution.

6. Conclusion and Strategic Recommendations

This research demonstrates that customer churn in the banking sector is not a random occurrence but a pattern-driven phenomenon. By analyzing 10,000 records, we have identified that geographic location (specifically the German market) and life-stage (ages 46–60) are the primary predictors of attrition.

Based on these findings, the following strategic interventions are recommended:

- **Localized German Strategy:** Conduct a deep-dive audit into the German market to understand if the 32% churn rate is driven by competitive pricing or specific service gaps.
- **Targeted Mature-Segment Retention:** Develop specialized loyalty programs for customers aged 45+ to protect high-value, long-term accounts.
- **Product Ecosystem Optimization:** Investigate the high failure rate of multi-product bundles. Simplifying the user experience for customers with 3 or more products could significantly reduce churn.

EXECUTIVE MEMORANDUM

TO: Department of Financial Oversight / Banking Regulatory Stakeholders

FROM: Haritha Mohan, Data Analytics Intern

DATE: May 12, 2026

SUBJECT: Strategic Assessment of Consumer Attrition and Market Stability

1. Objective This memorandum provides a high-level summary of customer attrition (churn) patterns within the European retail banking sector. The goal is to highlight systemic risks and demographic vulnerabilities that impact broader financial stability and consumer protection.

2. Critical Risk Indicators

- **Regional Instability:** A significant churn outlier was identified in the German market (32.44%). Such high volatility in a major economy warrants further investigation into localized competitive pressures or service gaps that may destabilize regional banking loyalty.
- **Demographic Vulnerability:** Citizens in the 46–60 age bracket exhibit the highest rates of attrition. As this group typically represents the most financially established segment of the public, their exodus suggests a need for more stable, long-term wealth-preservation products.
- **Consumer Stress Points:** Data indicates a near 100% exit rate for consumers holding 3 or 4 banking products. This suggests that "Product Saturation" and the resulting complexity are major drivers of consumer dissatisfaction.

3. Strategic Recommendations for Stakeholders

- **Simplify Consumer Portfolios:** Regulatory encouragement of simplified product structures could reduce consumer burnout and improve long-term institutional stability.
- **Targeted Engagement:** Banks should be encouraged to develop specialized retention frameworks for mature demographics to ensure they are not underserved by digital-only transitions.
- **Market Transparency:** Continued use of transparent data analytics—like the dashboard developed for this project—is essential for objective market oversight.